

Complete Derivation of the Quadrotor Equations of Motion

From rotor forces to a state-space model on $\mathbb{R}^3 \times \text{SO}(3)$

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Abstract

This note derives the rigid-body equations used in the UAV Motion Planning talk. The convention is chosen to match the slide exactly:

$$m\ddot{\mathbf{p}} = fRe_3 - mge_3, \quad \dot{R} = R\hat{\omega}, \quad J\dot{\omega} = \tau - \omega \times J\omega.$$

Here R maps body-frame coordinates into inertial-frame coordinates, ω is expressed in the body frame, and positive thrust acts along the body $+e_3$ axis. If a vehicle or text uses NED coordinates or thrust along $-e_3$, the corresponding signs must be changed consistently.

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1 Modeling assumptions and notation

We model the airframe as one rigid body with constant mass m and constant body-frame inertia matrix $J = J^\top \succ 0$. The center of mass is the body-frame origin. Rotors produce axial thrust and reaction torque; blade flapping, motor dynamics, ground effect, and aerodynamic drag are initially omitted and added later as extensions.

There are two right-handed frames:

- $\mathcal{I}$: inertial/world frame, with $\mathbf{e}_3 = [0, 0, 1]^\top$ pointing upward;
- $\mathcal{B}$: body frame fixed to the vehicle, also with body thrust along $+\mathbf{e}_3$.

The state variables and inputs are

$$\mathbf{p}, \mathbf{v} \in \mathbb{R}^3, \quad R \in \text{SO}(3), \quad \boldsymbol{\omega} \in \mathbb{R}^3, \quad f \geq 0, \quad \boldsymbol{\tau} \in \mathbb{R}^3,$$

where $\mathbf{p}$ and $\mathbf{v}$ are inertial-frame position and velocity, R maps a body-coordinate vector into inertial coordinates, and $\boldsymbol{\omega}$ is body angular velocity expressed in $\mathcal{B}$. A rotation matrix satisfies

$$R^\top R = I, \quad \det R = 1, \quad \mathbf{x}_{\mathcal{I}} = R\mathbf{x}_{\mathcal{B}}.$$

2 The hat operator and cross products

For $\mathbf{a} = [a_x, a_y, a_z]^\top$, define

$$\hat{\mathbf{a}} := \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix}.$$

Direct multiplication gives $\hat{\mathbf{a}}\mathbf{b} = \mathbf{a} \times \mathbf{b}$ for every $\mathbf{b} \in \mathbb{R}^3$. The matrix is skew-symmetric:

$$\hat{\mathbf{a}}^\top = -\hat{\mathbf{a}}.$$

Thus $\hat{\mathbf{a}}$ is the linear operator “cross with $\mathbf{a}$.” This identity is the key to writing angular-velocity kinematics as a matrix differential equation.

3 Translational dynamics from Newton’s law

Newton’s second law in the inertial frame is

$$m\ddot{\mathbf{p}} = \sum \mathbf{F}_{\mathcal{I}}.$$

Gravity acts downward:

$$\mathbf{F}_g = -mg\mathbf{e}_3.$$

Each rotor pushes along the body thrust axis. If rotor i produces magnitude $f_i \geq 0$, its body-frame force is $f_i\mathbf{e}_3$. All four rotor forces are parallel, so their total body-frame force is

$$\mathbf{F}_{T,\mathcal{B}} = \sum_{i=1}^4 f_i\mathbf{e}_3 = f\mathbf{e}_3, \quad f := \sum_{i=1}^4 f_i.$$

Because R maps body vectors into the inertial frame,

$$\mathbf{F}_{T,\mathcal{I}} = R\mathbf{F}_{T,\mathcal{B}} = fR\mathbf{e}_3.$$

Adding thrust and gravity yields

$$\boxed{m\ddot{\mathbf{p}} = fR\mathbf{e}_3 - mg\mathbf{e}_3.} \quad (1)$$

Introducing $\mathbf{v} = \dot{\mathbf{p}}$ gives

$$\dot{\mathbf{p}} = \mathbf{v}, \quad \dot{\mathbf{v}} = \frac{f}{m}R\mathbf{e}_3 - g\mathbf{e}_3.$$

This equation explains underactuation. The vehicle directly commands only the scalar f along its current body axis $R\mathbf{e}_3$; lateral acceleration requires changing R first.

3.1 Hover and acceleration feasibility

At level hover, $R = I$ and $\ddot{\mathbf{p}} = 0$, so $f = mg$. More generally, for a requested acceleration $\mathbf{a}_d$,

$$fR\mathbf{e}_3 = m(\mathbf{a}_d + g\mathbf{e}_3).$$

Therefore the required thrust direction and magnitude are

$$\mathbf{b}_{3,d} = R_d\mathbf{e}_3 = \frac{\mathbf{a}_d + g\mathbf{e}_3}{\|\mathbf{a}_d + g\mathbf{e}_3\|}, \quad f_d = m\|\mathbf{a}_d + g\mathbf{e}_3\|.$$

This is the central bridge from a planned position trajectory to a feasible attitude and thrust command.

4 Attitude kinematics and the rotation-matrix differential equation

Take any vector $\mathbf{r}$ fixed in the body. Its body coordinates are constant, while its inertial coordinates are

$$\mathbf{r}_{\mathcal{I}} = R\mathbf{r}_{\mathcal{B}}.$$

Differentiating gives

$$\dot{\mathbf{r}}_{\mathcal{I}} = \dot{R}\mathbf{r}_{\mathcal{B}}.$$

Rigid-body rotation says that the inertial velocity of this point is angular velocity crossed with its displacement. Because $\boldsymbol{\omega}$ and $\mathbf{r}_{\mathcal{B}}$ are both expressed in the body frame,

$$\dot{\mathbf{r}}_{\mathcal{I}} = R(\boldsymbol{\omega} \times \mathbf{r}_{\mathcal{B}}) = R\hat{\boldsymbol{\omega}}\mathbf{r}_{\mathcal{B}}.$$

This equality holds for every body vector $\mathbf{r}_{\mathcal{B}}$; hence the linear maps multiplying it must be equal:

$$\boxed{\dot{R} = R\hat{\boldsymbol{\omega}}.} \quad (2)$$

The same result follows from the rotation constraint. Differentiate $R^\top R = I$:

$$\dot{R}^\top R + R^\top \dot{R} = 0.$$

Thus $R^\top \dot{R}$ is skew-symmetric, so there is a unique vector $\boldsymbol{\omega}$ such that

$$R^\top \dot{R} = \hat{\boldsymbol{\omega}},$$

which again gives $\dot{R} = R\hat{\boldsymbol{\omega}}$. This also proves that the differential equation preserves orthogonality:

$$\frac{d}{dt}(R^\top R) = \hat{\boldsymbol{\omega}}^\top + \hat{\boldsymbol{\omega}} = 0.$$

Spatial-angular-velocity convention. If angular velocity is instead expressed in the inertial frame as $\boldsymbol{\Omega} = R\boldsymbol{\omega}$, then

$$\dot{R} = \widehat{\boldsymbol{\Omega}}R.$$

The order changes because the coordinate convention changes.

5 Rotational dynamics from angular momentum balance

The body-frame inertia tensor is

$$J = \int_{\mathcal{B}} \left(\|\mathbf{r}\|^2 I - \mathbf{r}\mathbf{r}^\top \right) dm,$$

and body-frame angular momentum is

$$\mathbf{h}_{\mathcal{B}} = J\boldsymbol{\omega}.$$

Its inertial representation is $\mathbf{h}_{\mathcal{I}} = RJ\boldsymbol{\omega}$. Differentiate using $\dot{R} = R\widehat{\boldsymbol{\omega}}$:

$$\begin{aligned} \dot{\mathbf{h}}_{\mathcal{I}} &= \dot{R}J\boldsymbol{\omega} + RJ\dot{\boldsymbol{\omega}} \\ &= R\widehat{\boldsymbol{\omega}}J\boldsymbol{\omega} + RJ\dot{\boldsymbol{\omega}} \\ &= R(\boldsymbol{\omega} \times J\boldsymbol{\omega} + J\dot{\boldsymbol{\omega}}). \end{aligned}$$

The angular-momentum theorem says $\dot{\mathbf{h}}_{\mathcal{I}} = \boldsymbol{\tau}_{\mathcal{I}}$. With body torque $\boldsymbol{\tau}$ and $\boldsymbol{\tau}_{\mathcal{I}} = R\boldsymbol{\tau}$,

$$R\boldsymbol{\tau} = R(J\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times J\boldsymbol{\omega}).$$

Multiplying by $R^\top$ gives Euler's rigid-body equation:

$$\boxed{J\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times J\boldsymbol{\omega} = \boldsymbol{\tau}.} \quad (3)$$

Equivalently,

$$\boxed{J\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times J\boldsymbol{\omega}, \quad \dot{\boldsymbol{\omega}} = J^{-1}(\boldsymbol{\tau} - \boldsymbol{\omega} \times J\boldsymbol{\omega}).} \quad (4)$$

The term $\boldsymbol{\omega} \times J\boldsymbol{\omega}$ is the gyroscopic coupling created by expressing angular momentum in a rotating body frame. If $J = \text{diag}(J_x, J_y, J_z)$, the component equations are

$$\begin{aligned} J_x\dot{\omega}_x &= \tau_x - (J_z - J_y)\omega_y\omega_z, \\ J_y\dot{\omega}_y &= \tau_y - (J_x - J_z)\omega_z\omega_x, \\ J_z\dot{\omega}_z &= \tau_z - (J_y - J_x)\omega_x\omega_y. \end{aligned}$$

6 From four rotor speeds to collective thrust and body torque

For rotor speed $\Omega_i \geq 0$, the common static model is

$$f_i = k_f \Omega_i^2, \quad q_i = s_i k_m \Omega_i^2,$$

where $k_f > 0$ is the thrust coefficient, $k_m > 0$ is the reaction-torque coefficient, and $s_i \in \{+1, -1\}$ records rotor spin direction. If rotor i is at body position $\mathbf{r}_i$, then

$$\mathbf{f} = \sum_{i=1}^4 f_i \mathbf{e}_3, \quad \boldsymbol{\tau} = \sum_{i=1}^4 (\mathbf{r}_i \times f_i \mathbf{e}_3 + q_i \mathbf{e}_3).$$

The first torque is the moment arm crossed with thrust; the second is aerodynamic reaction torque about the rotor axis.

For a “plus” configuration with

$$\mathbf{r}_1 = \ell \mathbf{e}_1, \quad \mathbf{r}_2 = \ell \mathbf{e}_2, \quad \mathbf{r}_3 = -\ell \mathbf{e}_1, \quad \mathbf{r}_4 = -\ell \mathbf{e}_2,$$

and spin signs $(s_1, s_2, s_3, s_4) = (+1, -1, +1, -1)$,

$$\begin{bmatrix} f \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} k_f & k_f & k_f & k_f \\ 0 & \ell k_f & 0 & -\ell k_f \\ -\ell k_f & 0 & \ell k_f & 0 \\ k_m & -k_m & k_m & -k_m \end{bmatrix} \begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix}.$$

Changing rotor numbering, choosing an X configuration, reversing a spin convention, or taking thrust along $-\mathbf{e}_3$ changes this allocation matrix but not the rigid-body equations.

7 Complete nonlinear state-space model

Collect the state and control as

$$\mathbf{x} = (\mathbf{p}, \mathbf{v}, R, \boldsymbol{\omega}) \in \mathbb{R}^3 \times \mathbb{R}^3 \times \text{SO}(3) \times \mathbb{R}^3, \quad \mathbf{u} = (f, \boldsymbol{\tau}).$$

The full ideal rigid-body model is

$$\boxed{\begin{aligned} \dot{\mathbf{p}} &= \mathbf{v}, \\ \dot{\mathbf{v}} &= \frac{f}{m} R \mathbf{e}_3 - g \mathbf{e}_3, \\ \dot{R} &= R \hat{\boldsymbol{\omega}}, \\ \dot{\boldsymbol{\omega}} &= J^{-1}(\boldsymbol{\tau} - \boldsymbol{\omega} \times J \boldsymbol{\omega}). \end{aligned}} \tag{5}$$

The model has twelve degrees of freedom locally (three position, three velocity, three attitude, and three angular velocity) but only four independent control inputs. It is nonlinear because thrust direction depends on R and because the rotational dynamics contain products of angular-velocity components.

8 Euler-angle and quaternion forms

The rotation-matrix form is globally nonsingular and is preferred for derivation. For roll–pitch–yaw angles with

$$R = R_z(\psi) R_y(\theta) R_x(\phi),$$

body angular velocity satisfies

$$\boldsymbol{\omega} = \begin{bmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \phi & \sin \phi \cos \theta \\ 0 & -\sin \phi & \cos \phi \cos \theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}.$$

Inverting this map introduces $\tan \theta$ and $\sec \theta$, exposing the pitch singularity at $\theta = \pm\pi/2$.

For a unit quaternion q representing the same body-to-inertial rotation and body angular velocity $\boldsymbol{\omega}$,

$$\dot{q} = \frac{1}{2}q \otimes \begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix}, \quad \|q\| = 1,$$

with the precise multiplication side/sign depending on quaternion storage and active/passive conventions. The translational and Euler equations are unchanged after replacing R by $R(q)$.

9 Useful checks and common sign errors

1. **Hover check:** $R = I$, $f = mg$, $\boldsymbol{\tau} = 0$, and $\boldsymbol{\omega} = 0$ must give zero acceleration.
2. **Rotation-manifold check:** $\dot{R} = R\hat{\boldsymbol{\omega}}$ must preserve $R^\top R = I$.
3. **Torque-free energy check:** with $\boldsymbol{\tau} = 0$,

$$\frac{d}{dt} \left(\frac{1}{2} \boldsymbol{\omega}^\top J \boldsymbol{\omega} \right) = \boldsymbol{\omega}^\top J \dot{\boldsymbol{\omega}} = -\boldsymbol{\omega}^\top (\boldsymbol{\omega} \times J \boldsymbol{\omega}) = 0.$$

4. **Frame check:** never add vectors expressed in different frames. $f\mathbf{e}_3$ is a body vector; $fR\mathbf{e}_3$ is the same force in inertial coordinates.
5. **Convention check:** many aerospace texts use NED coordinates and body $+z$ downward, producing equations such as $m\ddot{\mathbf{p}} = mge_3 - fRe_3$. This is not a disagreement in physics, only a different convention.

10 Extensions used in higher-fidelity models

The ideal equations become

$$\begin{aligned} m\dot{\mathbf{v}} &= fR\mathbf{e}_3 - mge_3 + \mathbf{F}_{\text{drag}} + \mathbf{F}_{\text{wind}}, \\ J\dot{\boldsymbol{\omega}} &= \boldsymbol{\tau} - \boldsymbol{\omega} \times J\boldsymbol{\omega} + \boldsymbol{\tau}_{\text{aero}} + \boldsymbol{\tau}_{\text{gyro}}. \end{aligned}$$

Common additions include linear or quadratic drag, rotor inflow, motor first-order dynamics, blade flapping, gyroscopic rotor momentum, ground effect, and time-varying mass. These terms improve fidelity but do not change the frame bookkeeping or the Newton–Euler derivation above.

11 Final result

Starting from Newton’s force balance, rigid-body kinematics, and angular-momentum balance gives

$$\begin{aligned} m\ddot{\mathbf{p}} &= fR\mathbf{e}_3 - mge_3, \\ \dot{R} &= R\hat{\boldsymbol{\omega}}, \\ J\dot{\boldsymbol{\omega}} &= \boldsymbol{\tau} - \boldsymbol{\omega} \times J\boldsymbol{\omega}. \end{aligned}$$

The rotor allocation map converts four squared motor speeds into the collective thrust and body torque appearing here. These equations are the dynamical feasibility constraint behind quadrotor motion planning: a collision-free curve is flyable only when some admissible $f(t)$, $\boldsymbol{\tau}(t)$, and $R(t)$ realize it.